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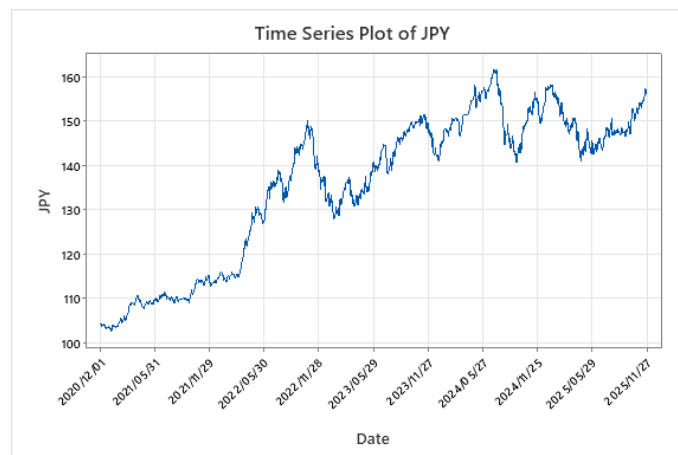
Forecasting Time Series Data, STAT-GB 6018

8 December, 2025

FORECASTING PROJECT 2

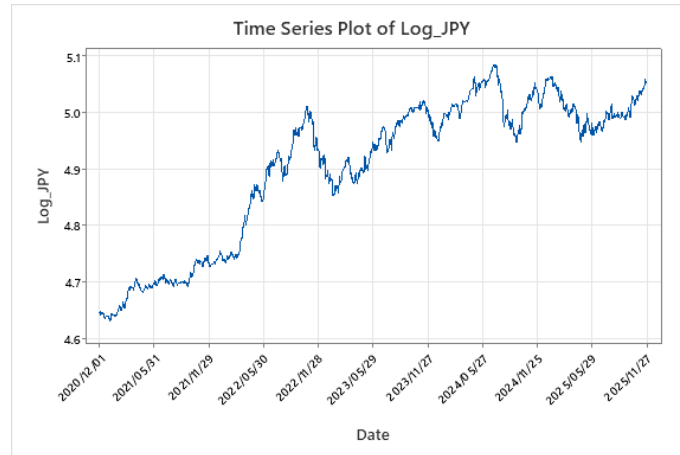
This project aims to analyze, model, and forecast the daily USD/JPY exchange rate by applying the ARIMA-ARCH framework. The data are obtained from **Yahoo Finance** under the ticker JPY=X, which reports the **adjusted closing prices of the Japanese yen per U.S. dollar**. The series is recorded at a **daily frequency** and spans the period from December 1, 2020 through November 28, 2025. Before specifying an appropriate model, I conducted an exploratory analysis of the raw exchange-rate series to examine its overall dynamics, variance behavior, and potential non-stationarity.

To begin, I plotted the time series plot of JPY to visualize its general pattern.

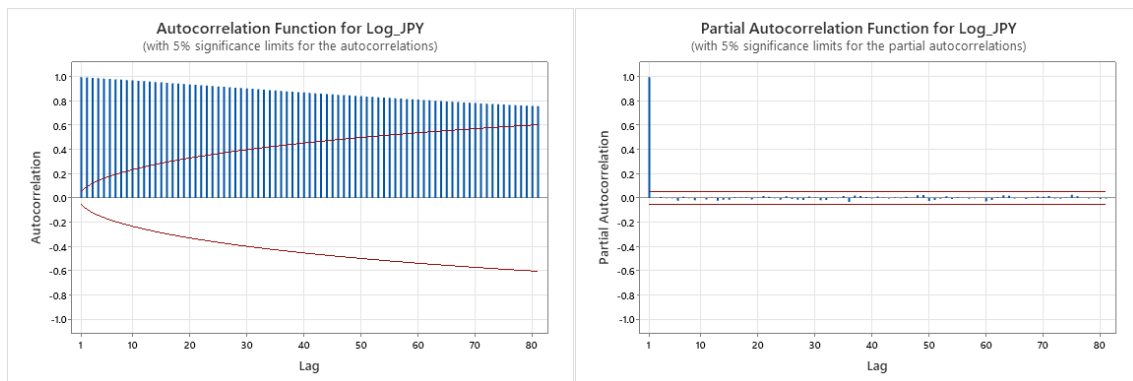


The series showed a clear upward trend with a seeming conditional volatility, resembling exponential growth. To stabilize the variance and smooth the model, we should convert the exponential trend growth into a linear trend. Thus, I decided to take the logarithm of the data to achieve this. At the same time, I plotted the time series of its ACF and PACF plots.

After taking log, the time series plot of Log_JPY and its ACF & PACF are shown as follows:



Time Series Plot of Log_JPY

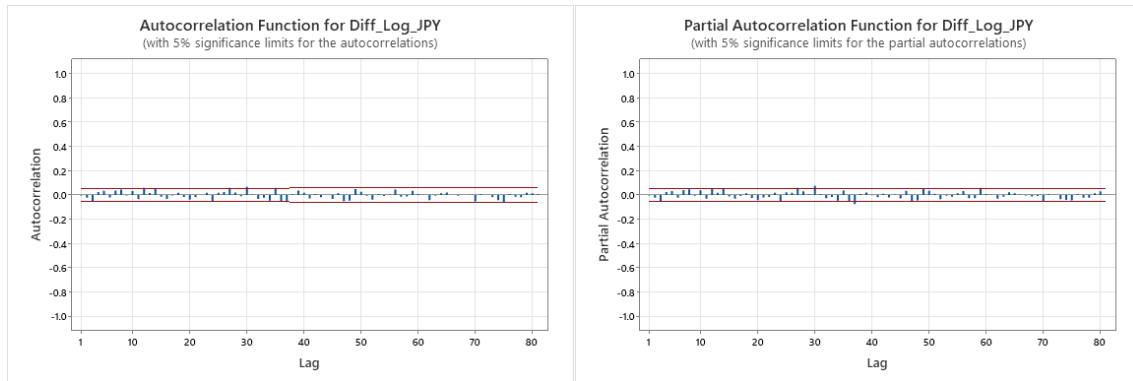


ACF of Log_JPY

PACF of Log_JPY

By observing the plots, the ACF of the Log_JPY series declines extremely slowly, with autocorrelation values remaining strongly positive across many lags. Such behavior is typical of a **non-stationary** time series. In addition, the PACF of the Log Rupee series shows a value of 1 at the first lag, which is a significant sign of random walk. Summing up, the series should be non-stationary.

To achieve stationarity, I differenced the Log_JPY series and then evaluated the resulting ACF and PACF.



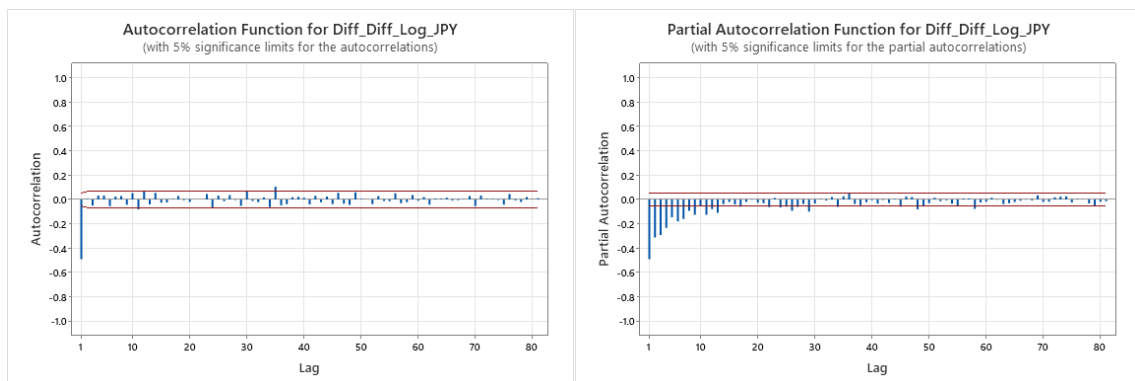
ACF of Diff_Log_JPY

PACF of Diff_Log_JPY

After taking the first difference, a small negative autocorrelation appears at lag 3 and lag 8 in both the ACF and PACF, while nearly all remaining autocorrelations lie comfortably within the 5% significance limits and show no structured pattern. This pattern suggests that the Diff_Log_JPY series is stationary.

From these plots, a plausible model choice is an $ARIMA(3,d,3)$.

To verify this selection, I applied a second differencing to determine the d in the ARIMA model. The ACF and PACF plots for the differenced Diff_Log_JPY series are displayed below:



ACF of 2nd Diff_Log_JPY

PACF of 2nd Diff_Log_JPY

The plots both show a strong negative autocorrelation at lag 1, which means that it is overdifferenced, removing meaningful long-term structure and increasing noise. To sum up, the series required a first difference ($d = 1$) to achieve stationarity.

From the preceding analysis, an $ARIMA(3,1,3)$ model can be identified as a feasible model.

To identify the most appropriate ARIMA model for the differenced Log series, I estimated several ARIMA models with $d=1$ and within the range $0 \leq p \leq 5$ and $0 \leq q \leq 5$, and compared their AICc. Among the candidate models, the one with the lowest AICc value was selected as the best ARIMA model for the Log dataset.

Model Selection

Model (d = 1)	LogLikelihood	AICc	AIC	BIC
p = 5, q = 4*	4820.69	-9619.17	-9619.37	-9562.50
p = 4, q = 4	4815.50	-9612.86	-9613.00	-9566.47
p = 2, q = 4	4813.25	-9610.39	-9610.50	-9569.14
p = 0, q = 0	4806.99	-9609.97	-9609.98	-9599.64
p = 3, q = 5	4813.56	-9608.97	-9609.11	-9562.58
p = 3, q = 4	4813.38	-9608.62	-9608.76	-9562.22
p = 5, q = 3	4814.35	-9608.53	-9608.70	-9557.00
p = 0, q = 1	4807.00	-9607.99	-9608.01	-9592.50
p = 1, q = 0	4807.00	-9607.99	-9608.00	-9592.49
p = 4, q = 1	4810.99	-9607.90	-9607.98	-9571.79
p = 3, q = 0	4808.95	-9607.85	-9607.89	-9582.04
p = 0, q = 3	4808.88	-9607.72	-9607.76	-9581.91
p = 2, q = 3	4810.53	-9606.97	-9607.06	-9570.87
p = 0, q = 4	4809.38	-9606.69	-9606.76	-9575.74
p = 0, q = 2	4807.35	-9606.67	-9606.70	-9586.02
p = 1, q = 5	4811.39	-9606.67	-9606.78	-9565.42
p = 2, q = 0	4807.35	-9606.67	-9606.70	-9586.02
p = 0, q = 5	4810.35	-9606.60	-9606.69	-9570.50
p = 4, q = 0	4809.32	-9606.57	-9606.64	-9575.62
p = 1, q = 3	4809.07	-9606.07	-9606.13	-9575.11
p = 4, q = 2	4811.01	-9605.90	-9606.01	-9564.65
p = 5, q = 0	4809.98	-9605.88	-9605.96	-9569.77
p = 2, q = 2	4808.77	-9605.47	-9605.54	-9574.52
p = 1, q = 2	4807.69	-9605.34	-9605.38	-9579.53
p = 2, q = 5	4811.62	-9605.10	-9605.24	-9558.71
p = 1, q = 1	4805.40	-9604.78	-9604.79	-9589.28
p = 5, q = 1	4810.34	-9604.56	-9604.68	-9563.32
p = 4, q = 3	4811.22	-9604.30	-9604.44	-9557.90
p = 5, q = 2	4811.19	-9604.24	-9604.38	-9557.85
p = 3, q = 3	4808.64	-9603.20	-9603.28	-9567.09
p = 2, q = 1	4805.57	-9603.11	-9603.14	-9582.46
p = 1, q = 4	4806.86	-9601.65	-9601.72	-9570.70

* Best model with minimum AICc.

Model Selection

Model (d = 1)	LogLikelihood	AICc	AIC	BIC
p = 5, q = 4*	4819.09	-9618.01	-9618.18	-9566.48
p = 5, q = 5	4819.35	-9616.49	-9616.70	-9559.83
p = 4, q = 4	4815.50	-9612.86	-9613.00	-9566.47
p = 3, q = 5	4813.56	-9608.97	-9609.11	-9562.58
p = 2, q = 4	4811.42	-9608.75	-9608.84	-9572.65
p = 0, q = 0	4805.24	-9608.48	-9608.48	-9603.31
p = 5, q = 3	4813.06	-9607.98	-9608.12	-9561.59
p = 4, q = 1	4809.71	-9607.36	-9607.43	-9576.41
p = 3, q = 4	4811.53	-9606.95	-9607.07	-9565.70
p = 0, q = 1	4805.24	-9606.48	-9606.49	-9596.15
p = 1, q = 0	4805.24	-9606.48	-9606.49	-9596.15
p = 3, q = 0	4806.92	-9605.81	-9605.84	-9585.16
p = 0, q = 3	4806.87	-9605.70	-9605.73	-9585.05
p = 2, q = 2	4807.77	-9605.50	-9605.55	-9579.70
p = 4, q = 2	4809.73	-9605.37	-9605.45	-9569.26
p = 2, q = 3	4808.62	-9605.17	-9605.24	-9574.22
p = 0, q = 5	4808.55	-9605.03	-9605.09	-9574.07
p = 2, q = 0	4805.51	-9605.01	-9605.03	-9589.52
p = 0, q = 2	4805.51	-9605.00	-9605.02	-9589.51
p = 0, q = 4	4807.46	-9604.87	-9604.91	-9579.06
p = 1, q = 1	4805.40	-9604.78	-9604.79	-9589.28
p = 4, q = 0	4807.39	-9604.74	-9604.79	-9578.94
p = 5, q = 0	4808.19	-9604.31	-9604.37	-9573.35
p = 1, q = 3	4807.07	-9604.10	-9604.14	-9578.29
p = 4, q = 3	4809.93	-9603.75	-9603.86	-9562.50
p = 5, q = 2	4809.91	-9603.70	-9603.81	-9562.45
p = 1, q = 5	4808.82	-9603.56	-9603.65	-9567.46
p = 1, q = 2	4805.77	-9603.51	-9603.54	-9582.86
p = 3, q = 3	4808.64	-9603.20	-9603.28	-9567.09
p = 2, q = 1	4805.57	-9603.11	-9603.14	-9582.46
p = 5, q = 1	4808.52	-9602.95	-9603.04	-9566.84
p = 1, q = 4	4806.86	-9601.65	-9601.72	-9570.70

* Best model with minimum AICc.

AICc with Constant

AICc without Constant

The comparison results are summarized above. With no constant, the best model is ARIMA (5, 1, 4) with AICc = -9618.01. With constant, the best model is ARIMA (5, 1, 4) with AICc = -9619.17, which is smaller than -1330.7. Therefore, the best-fitting model based on AICc is **ARIMA (5, 1, 4) with constant term**.

The estimated ARIMA(5, 1, 4) model is given by the estimates of parameters as follows:

Final Estimates of Parameters

Type	Coef	SE Coef	T-Value	P-Value
AR 1	0.3709	0.0450	8.23	0.000
AR 2	0.7958	0.0432	18.41	0.000
AR 3	0.3263	0.0465	7.01	0.000
AR 4	-0.8859	0.0214	-41.35	0.000
AR 5	0.0384	0.0288	1.33	0.183
MA 1	0.3822	0.0354	10.79	0.000
MA 2	0.8065	0.0337	23.93	0.000
MA 3	0.3669	0.0335	10.97	0.000
MA 4	-0.9295	0.0110	-84.41	0.000
Constant	0.000111	0.000061	1.81	0.070

Differencing: 1 Regular

Number of observations after differencing: 1300

Therefore, the estimated model should be:

$$\Delta \hat{x}_t = 0.3709\Delta \hat{x}_{t-1} + 0.7958\Delta \hat{x}_{t-2} + 0.3263\Delta \hat{x}_{t-3} - 0.8859\Delta \hat{x}_{t-4} + 0.0384\Delta \hat{x}_{t-5} + \hat{\epsilon}_t \\ - 0.3822\hat{\epsilon}_{t-1} - 0.8065\hat{\epsilon}_{t-2} - 0.3669\hat{\epsilon}_{t-3} + 0.9295\hat{\epsilon}_{t-4} + 0.000111$$

$$\hat{x}_t = \hat{x}_{t-1} + 0.3709(x_{t-1} - x_{t-2}) + 0.7958(x_{t-2} - x_{t-3}) + 0.3263(x_{t-3} - x_{t-4}) - 0.8859(x_{t-4} \\ - x_{t-5}) + 0.0384(x_{t-5} - x_{t-6}) + \hat{\epsilon}_t - 0.3822\hat{\epsilon}_{t-1} - 0.8065\hat{\epsilon}_{t-2} - 0.3669\hat{\epsilon}_{t-3} \\ + 0.9295\hat{\epsilon}_{t-4} + 0.000111$$

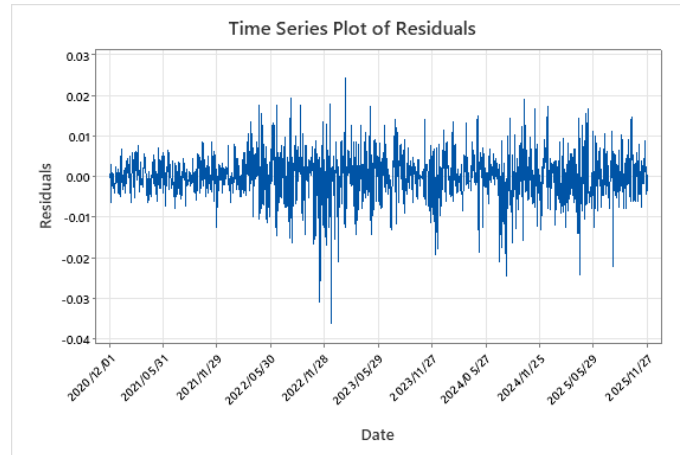
Also, the one step ahead forecast and 95% forecast interval calculated by Minitab is shown as follows:

Forecasts from Time Period 1301

Time Period	Forecast	SE Forecast	95% Limits		Actual
			Lower	Upper	
1302	5.05312	0.0059552	5.04145	5.06480	

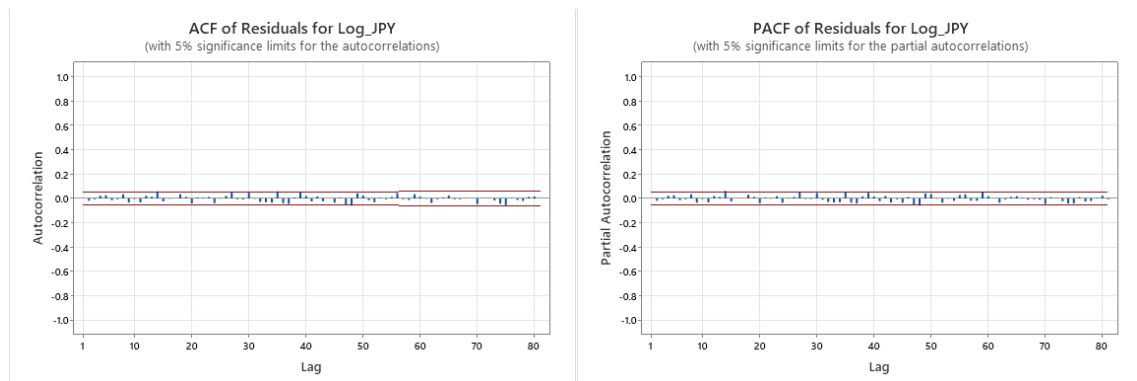
The one step ahead forecast is **5.05312**, and 95% forecast interval is **[5.04145, 5.06480]**.

Next, I generated a time-series plot of the residuals from the $ARIMA(5,1,4)$ model (with a constant term).



The time-series plot of the residuals shows that they fluctuate around zero. However, the magnitude of these fluctuations is not constant over time. Several periods show more obvious volatility compared with surrounding periods.

To evaluate whether the residuals exhibit any autocorrelation, I examined the ACF and PACF of the residual series.



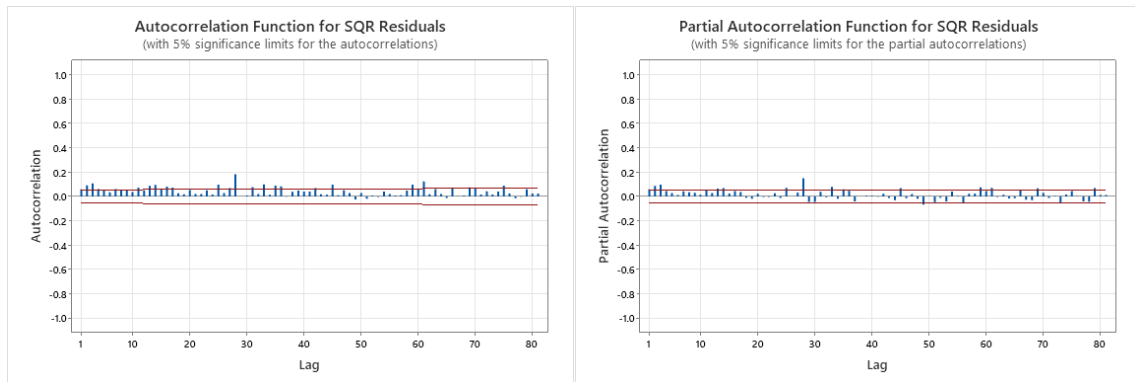
ACF of Residuals

PACF of Residuals

We can see that the residual fluctuates randomly around zero with no visible trend or structure, which ensure that systematic patterns have been removed. As for ACF and PACF plots, they show no significant spikes, and almost all correlations fall within the 95% confidence bounds. This pattern represents that the residuals are approximate white noise, meaning that the residuals behave as uncorrelated.

But uncorrelatedness alone does not guarantee independence. In order to check the independence, I plotted the ACF and PACF of squared residuals. If the residuals are

independent, there should be no statistically significant autocorrelations in both plots too. The plots are shown below:



ACF of Squared Residuals

PACF of Squared Residuals

The plots showed that the ACF and PACF of squared residuals display some significant spikes out of the 95% confidence bounds at several lags. This represents that the squared residuals are correlated with some of their own past values, which means that the condition:

$$Cov(\varepsilon_t^2, \varepsilon_{t-k}^2) = 0 \text{ for all } k > 0$$

does not hold. Therefore, the residuals are **not independent**.

Using R, I fitted ARCH(q) models for q = 0 to 10. Their log-likelihood and AICC values are shown as follow:

q	Log likelihood	AIC_C
0	4820.947	-9639.891
1	4823.591	-9643.173
2	4839.386	-9672.754
3	4846.876	-9685.722
4	4855.936	-9701.826

5	4854.133	-9696.202
6	4852.419	-9690.751
7	4854.111	-9692.111
8	4852.180	-9686.221
9	4850.876	-9681.582
10	4849.662	-9677.119

It confirmed that the **ARCH(4)** model has the lowest AIC_C = -9701.826, which gave the conclusion that it is the most fitting model among all these ARCH(q) models based on the AIC_C values.

To check whether ARCH model or GARCH model is more appropriate for the data, I evaluate the log likelihood values and AIC_C values for the $GARCH(1,1)$ model subsequently:

GARCH	Log likelihood	AIC_C
	4903.205	-9800.391

Since the value of AIC_C for $GARCH(1,1)$ model (= -9800.391) is smaller than that of the $ARCH(4)$ model (= -9701.826), I identified that the **GARCH(1,1) model** is fitter for the given data.

R also reports the estimated parameters and their corresponding p-values:

```

Coefficient(s):
      Estimate Std. Error  t value Pr(>|t|)
a0 2.074e-07   9.302e-08    2.230   0.0257 *
a1 4.742e-02   5.483e-03    8.649  <2e-16 ***
b1 9.483e-01   6.309e-03   150.314  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

- $\hat{\omega} = 2.074 \times 10^{-7}$, $p = 0.0257$
- $\hat{\alpha} = 0.04742$, $p = 2 \times 10^{-16}$
- $\hat{\beta} = 0.9483$, $p = 2 \times 10^{-16}$

Based on the p-values calculated by R, $\hat{\alpha}$ and $\hat{\beta}$ are **statistically significant at the 0.1% level**, and $\hat{\omega}$ is **statistically significant at 1% level**. These results provide strong evidence that all estimated coefficients play an important role in the model.

To achieve weakly stationary, $\hat{\alpha} + \hat{\beta}$ should be smaller than 1 because $E[h_t]$ is limit only if $\hat{\alpha} + \hat{\beta} < 1$. If $\hat{\alpha} + \hat{\beta} = 1$, the model is an I-GARCH. Here, $\hat{\alpha} + \hat{\beta} = 0.99572 < 1$, indicating that the process is stationary and does not correspond to an I-GARCH model.

Using all the estimated coefficients, the full expression of the $GARCH(1,1)$ model can be written as:

$$\hat{h}_t = 2,074 \times 10^{-7} + 0.04742\hat{h}_{t-1}^2 + 0.9483\hat{h}_{t-1}$$

The summary of final model selected and its logLik value is generated by the R, including the model, residuals, coefficients, diagnostic test and Box-Ljung test.

```

> summary(final_model)

Call:
garch(x = x, order = c(1, 1))

Model:
GARCH(1,1)

Residuals:
    Min       1Q   Median       3Q      Max
-5.2894 -0.5782  0.0487  0.5835  3.8235

Coefficient(s):
      Estimate Std. Error t value Pr(>|t|)
a0 2.074e-07   9.302e-08   2.230  0.0257 *
a1 4.742e-02   5.483e-03   8.649 <2e-16 ***
b1 9.483e-01   6.309e-03  150.314 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Diagnostic Tests:
      Jarque Bera Test

data: Residuals
X-squared = 203.76, df = 2, p-value < 2.2e-16

      Box-Ljung test

data: Squared.Residuals
X-squared = 0.21338, df = 1, p-value = 0.6441

> logLik(final_model)
'log Lik.' 4903.205 (df=3)

```

Also, I evaluated the unconditional variance of the shocks of this model:

$$Var[\varepsilon_t] = \frac{\hat{\omega}}{1 - \hat{\alpha} - \hat{\beta}} = \frac{2.074 \times 10^{-7}}{1 - 0.99572}$$

The R output of the unconditional variance of shock is **4.906391×10^{-5}** .

Using the Minitab output and the R output, I constructed the 95% one step ahead forecast interval for the Log_JPY value based on the selected *ARIMA*(5,1,4) - *GARCH*(1,1) model. By using R, I got the 2.5 percentile and the 97.5 percentile for the Log_JPY value.

```

> # Calculate the 95% forecast interval
> P2.5 <- 5.05312 + sigma_next * qnorm(0.025)
> P97.5 <- 5.05312 + sigma_next * qnorm(0.975)
> P2.5 <- as.numeric(P2.5)
> P97.5 <- as.numeric(P97.5)
> P2.5
[1] 5.044003
> P97.5
[1] 5.062237

```

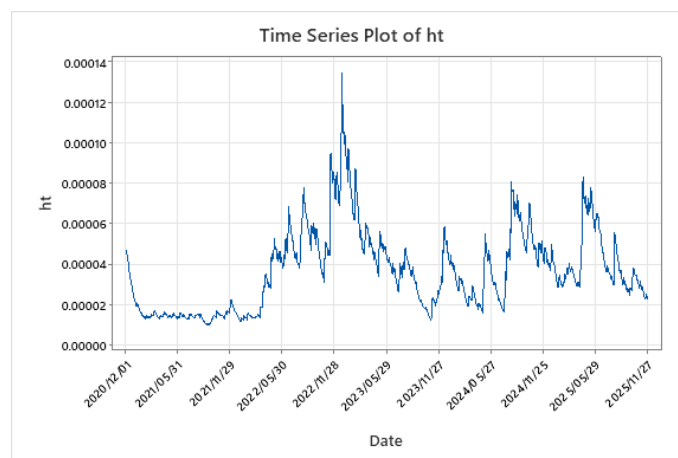
Hence, the 95% one step ahead forecast interval for the Log_JPY value is **[5.044003, 5.062238]**.

Compare with the interval [5.04145, 5.06480] based on the $ARIMA(5,1,4)$ only model from previous calculation, the interval based on the $ARIMA(5,1,4)$ - $GARCH(1,1)$ model for the Log_JPY value is **narrower**.

Also, by using R, I got the 5th percentile of the conditional distribution of the next period's Log_JPY value, which is **5.045469**.

```
> # Calculate the 5th percentile
> sigma_next <- sqrt(h_next)
> P5 <- 5.05312 + sigma_next * qnorm(0.05)
> P5 <- as.numeric(P5)
> P5
[1] 5.045469
```

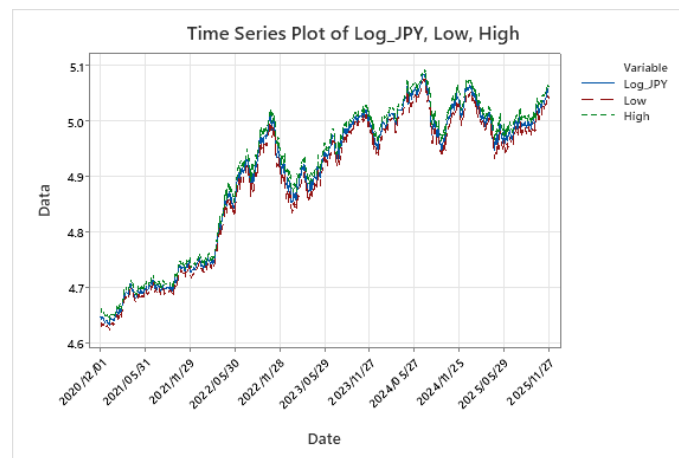
Then I plotted the conditional variances h_t for the $GARCH(1,1)$ model to locate the bursts of high volatility.



The plot highlighted that periods of high volatility occur intermittently: a obvious clustering emerges starting in 2022, culminating in a massive peak from late 2022 to 2023, followed by numerous smaller clusters until now. The conditional variance in the first half of the sequence exhibits minimal fluctuation, which shows far more stable variance.

These periods **agree with** what I observed from the time series plot of Log_JPY and the residual: within these specific time intervals, both the Log_JPY and residual time series plots exhibit more obvious fluctuations. This indicates that the selected $GARCH(1,1)$ model effectively captures the dynamic changes in volatility over time.

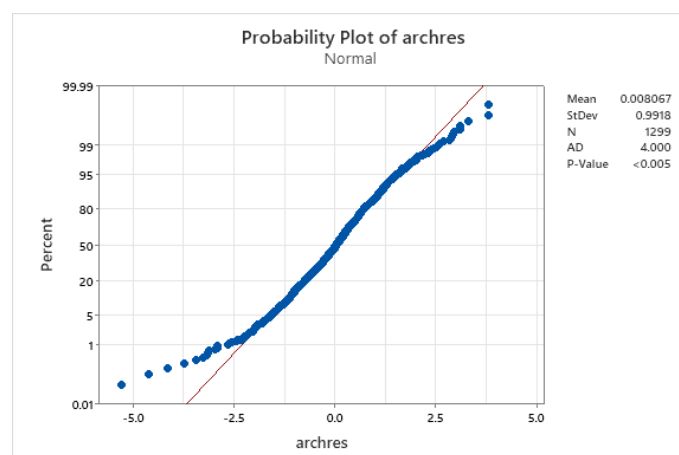
Next, I made the time series plot which simultaneously shows the Log_JPY and the ARIMA-ARCH one-step-ahead 95% forecast intervals.



From the time-series plot, the one-step-ahead 95% forecast intervals mostly included the actual Log_JPY values well. Most observations lie within the predicted area, suggesting that the model gives somehow accurate short-term forecasts. The width of intervals increased during periods of higher volatility and decreased when the series is more stable, which made the prediction more practical and reasonable.

However, on a few days when Log_JPY shows sudden movements, the real values approach or even went out of the forecast intervals. This shows that although the model adjusts to shifts in volatility, some sudden and extreme fluctuations are still difficult to predict precisely.

To further assess the adequacy of the ARIMA-ARCH model, I calculated the residuals and constructed a normal probability plot. Examining the residuals is essential because model diagnostics require checking whether the normality assumption is reasonable.



The normality test for the *archres* reveals clearly deviations in both tails, which means that the residuals **don't follow a perfect normal distribution**. Thus, the ARIMA-ARCH model **does not** fully account for the leptokurtosis (heavy-tailed) behavior present in the original data.

Using the criterion that whenever the absolute value of a residual exceeds 1.96, a 95% prediction interval is violated, I identified a total of **70 such failures**. The calculations are based on all 1301 observations, excluding two missing values (NAs), giving in an effective sample size of 1299. This gives a failure rate of about **5.388%**, which aligns well with the expectation for a 95% predictive interval.

From the JPY data obtained via Yahoo Finance, the adjusted closing price on December 1 is **155.928** with a log value of **5.04939**. According to the earlier forecast results, the 95% one-step-ahead interval from the ARMA-only model is [5.04145, 5.06480], while the interval obtained from the ARMA-ARCH model is [5.044003, 5.062238]. Both intervals successfully contain the real log price, suggesting that neither interval is too narrow. Each of them provides adequate coverage of the actual outcome. Meanwhile, since the observed value lies near the mid of both ranges, the intervals are not excessively wide. To sum up, the one-step-ahead prediction intervals from both the ARMA-only and ARMA-ARCH models seem to be **proper**.